

Muhammad Hilmy Putra Risma

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SUMMARY

Motivated and detail-oriented Mathematics graduate with a deep focus on data analytics and nearly three years of dedicated experience through academic projects and certified coursework. Skilled in data processing, visualization, and statistical modeling using Python, SQL, Excel, and Tableau. Passionate about the integration of mathematics, IT, and business to drive impactful, data-driven decisions and contribute meaningfully to real-world problem-solving.

SKILLS

- Data Analysis & Machine Learning:** Python with libraries like SciPy, Pandas, Matplotlib, Scikit-Learn, TensorFlow, Keras.
- Data Management & Querying:** Excel, PostgreSQL, MySQL, Pentaho, Talend, DBeaver, Google BigQuery, Airflow.
- Data Visualization & BI Tools:** Tableau, Metabase, Looker Studio, PowerBI.
- AWS Platform:** S3, Glue, Redshift, Athena, SageMaker, QuickSight, Kinesis, Step Functions, Lambda.

EXPERIENCE

Data Analytics | PT Bank Rakyat Indonesia (Persero) Tbk · Internship Dec 2025 – Present

- Performance Analysis:** Conducted employee productivity analysis across 18 Regional Offices by examining key performance indicators and demographic factors to identify workforce performance patterns and trends.
- Dashboard Development:** Developed productivity and employee profiling dashboards based on data from 26,000+ employees to support performance monitoring and management decision-making.
- Data Processing & NLP Analysis:** Processed and prepared large-scale datasets based on requests from cross-functional divisions, and performed survey data analysis using NLP techniques to generate insights across 4 main aspects for evaluation and decision-making support.

Data Engineer | AnterAja · Internship Jul – Dec 2025

- Data Management & Querying:** Managed and explored large-scale datasets using PostgreSQL and DBeaver, ensuring accurate data extraction and efficient query execution for business reporting needs.
- ETL Development:** Built and automated ETL pipelines using Pentaho Data Integration and Talend to clean, transform, and load raw data into structured formats across systems.
- Dashboard Development:** Created interactive dashboards in Metabase to monitor Job Orders, Cost, and Unit Utilization, providing real-time insights to support operational and strategic decisions.
- Cloud Integration:** Leveraged AWS services such as S3, Athena, and Lambda to support data processing workflows and improve scalability across pipeline operations.

Business Database Analyst | PT Pegadaian · Internship Aug – Dec 2023

- Data Collection & Analysis:** Utilized SQL, Excel, and Google BigQuery to gather and analyze financial data, providing actionable insights to support business marketing decisions.
- Data Visualization:** Developed interactive dashboards using Tableau and Excel, including a geographic potential map by analyzing customer distribution across 7 regions with population data, and a daily-updated KPI monitoring dashboard for 64 outlets to help branch managers track performance trends.
- Predictive Analysis:** Conducted a three-year analysis on Outstanding Loan (OSL) data and performed forecasting using the ARIMA time series model, achieving a forecast accuracy of 97%.

EDUCATION

- UNIVERSITY OF INDONESIA** Aug 2020 – Jan 2025
B.Sc. in Mathematics (Specialization: Data Sciences)
Emphasized Data Analysis, Statistical Modeling, Machine Learning, and Deep Learning

PROJECTS

- Comparative Evaluation of Multi-Large Language Model APIs for Sentiment Analysis** Apr 2025
Developed a sentiment analysis system in Python integrating DeepSeek, Gemini, and GPT APIs to evaluate 2,400+ text samples; conducted multi-model classification analyzing accuracy, response time, and API cost; visualized results with confusion matrices achieving up to 99% accuracy.
Keywords: Sentiment Analysis, Natural Language Processing , Large Language Models, Machine Learning
Repository: github.com/MuhammadHilmyPutraRisma/multi-llm-sentiment
- Comparison of In-Context Learning in GPT for Sentiment Analysis Level Topic** Jan 2025
Performed topic-level sentiment analysis on Tokocrypto app reviews by scraping data from the Play Store and App Store. Applied three in-context learning techniques using GPT models, with the best method achieving 99% accuracy. Utilized BERT for data representation and EFCM for topic clustering, identifying key topics with representative keywords for interpretation.
Keywords: Topic-Level Sentiment Analysis, In-Context Learning, GPT, BERT-EFCM
Repository: github.com/MuhammadHilmyPutraRisma/tokocrypto-sentiment
- Bitcoin Price Forecast with ARIMA Model** Jan 2023
Developed ARIMA models in Python and R to forecast Bitcoin prices seven days ahead. Achieved high accuracy with an average RMSE of under \$200. Effectively visualized and presented insights and predictions to academic assessors, demonstrating strong forecasting and communication skills.
Keywords: Time Series Forecasting, Statistical Modeling, ARIMA Model, Predictive Analytics
Repository: github.com/MuhammadHilmyPutraRisma/bitcoin-arima-forecast

CERTIFICATES

- AWS Cloud Quest: Machine Learning** – AWS Training & Certification, Nov 2025
- AWS SimuLearn: Data Analytics** – AWS Training & Certification, Sep 2025
- Deep Learning Specialization** – DeepLearning.AI, May 2025
- Machine Learning Specialization** – Stanford University, May 2025
- Mathematics for Machine Learning** – Imperial College London, Apr 2025
- Transforming Data Into Action** – RIS Consulting, Feb 2025